LAB 6: System Analysis & Planning

Subject: System Simulation Lab / Capstone Project

Submission Date: 23 August 2026

Project Title: Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data

Challenge Reference: MathWorks Excellence in Innovation Challenge — Project #255

Institution: Vidyalankar Institute of Technology (VIT), Mumbai

Department: Department of Electronics and Computer Science Engineering

Team Members & Contributors:

1. **Rajiv Gohil** (Project Lead, Powertrain Dynamics & Simulink, Route Engine) — `rajiv.gohil24@vit.edu.in`
2. **Sneha Pardeshi** (Google Maps API & Driving Cycles, Energy Accounting, Charging Optimization & UI) — `sneha.pardeshi@vit.edu.in`

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1. Project Overview & Motivation

As electric vehicles (EVs) become central to sustainable transportation, accurate trip planning and energy management are paramount. Standard homologation cycles (such as NEDC or WLTP) fail to capture real-world driving conditions, leading to unexpected range depletion and "range anxiety" for EV drivers. Real-world energy consumption is heavily dictated by:

- Topographical elevation variations (hill climbs and descents),
- Dynamic traffic bottlenecks and speed fluctuations,
- Speed limits and aerodynamic drag scaling quadratically with speed,
- Ambient temperature variations and auxiliary HVAC electrical loads.

This project designs a **simulation-based workflow for Battery Electric Vehicles (BEVs)** that couples real-time mapping data (via Google Maps API) with detailed longitudinal powertrain physics and equivalent circuit battery models in **MATLAB® and Simulink®**.

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2. System Requirements Specification

2.1 Functional Requirements (FR)

- **FR-1 (Real-Time Route Ingestion):** Interface MATLAB with Google Maps Directions & Elevation APIs to extract waypoint coordinates, route distance, step duration, legal speed limits, and continuous elevation profiles for any origin-destination pair.
- **FR-2 (Offline Fallback Capability):** Provide high-resolution pre-compiled datasets for key Indian highway corridors (e.g., Mumbai-Pune Expressway, Old NH48, Yamuna Expressway) ensuring zero-dependency operation.
- **FR-3 (Driving Cycle Synthesizer):** Synthesize a continuous 2 Hz time-speed trajectory *[t, v(t), \theta(t)]* reflecting traffic flow and road grade, formatted as a standard MATLAB `timeseries` and MAT-file for the Simulink **Driving Cycle Source block**.
- **FR-4 (BEV Modeling via Virtual Vehicle Composer):** Define customizable BEV models (curb mass, aerodynamic drag *C_d*, frontal area *A*, rolling resistance *C_{rr}*, motor envelope, and battery capacity) parameterized matching MathWorks Virtual Vehicle Composer.
- **FR-5 (Powertrain Simulation & Energy Accounting):** Accurately compute tractive forces, motor electrical losses, regenerative braking recovery, net energy consumption (kWh), efficiency (kWh/100 km), battery State of Charge (SOC)

trajectory, and operational electricity costs in INR (■).

- **FR-6 (Driving Modes Optimization):** Implement user-selectable driving modes (**Eco**, **Normal**, **Performance**) with dynamic acceleration constraints, cruising speed caps, regen prioritization, and HVAC power scaling.
- **FR-7 (Charging Stop Optimization):** Detect battery reserve boundary violations (*SOC < 15\%*), search for candidate DC fast charging stations along the route corridor, and compute CC-CV charging durations to 80% SOC.
- **FR-8 (Interactive Visualization & Reporting):** Render interactive maps (satellite imagery and roadmap tiles) with elevation profiles and provide a one-click PDF summary report export.

2.2 Non-Functional Requirements (NFR)

- **NFR-1 (Engineering Fidelity):** Energy estimations must align within *\pm 5\%* of empirical on-road BEV test data.
- **NFR-2 (Execution Speed):** Analytical route simulation solver must complete a 150 km corridor evaluation in under 3 seconds.
- **NFR-3 (Modularity & Maintainability):** Decoupled architecture separating Route Engine, Vehicle Database, Driving Cycle Generator, Powertrain Physics, and UI.

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3. Mathematical & Theoretical Modeling Framework

```
flowchart LR
    A["Road Coordinates & Elevation (Google Maps)"] --> B["Driving Cycle Synthesizer [t, v(t), grade]"]
    B --> C["Longitudinal Vehicle Dynamics (F_net = m*a)"]
    C --> D["Electric Motor Map (T, omega, Losses)"]
    D --> E["Battery ECM (Voc - I*Rint)"]
    E --> F["KPIs: SOC(t), Net kWh, Cost INR"]
```

3.1 Longitudinal Vehicle Dynamics

The dynamic force equilibrium along the vehicle's longitudinal axis is governed by Newton's Second Law:

$$m_{eff} \frac{dv}{dt} = F_{traction} - F_{rolling} - F_{aero} - F_{grade} - F_{frictionBrake}$$

Where:

1. **Aerodynamic Drag:**

$$F_{aero} = \frac{1}{2} \rho_{air} C_d A_{frontal} v^2$$

2. **Tire Rolling Resistance:**

$$F_{rolling} = C_{rr} m_{veh} g \cos(\theta) \tanh\left(\frac{v}{0.1}\right)$$

3. **Road Grade Force:**

$$F_{grade} = m_{veh} g \sin(\theta) \quad where \theta = \arctan\left(\frac{grade_{\%}}{100}\right)$$

4. **Effective Rotational Inertia:**

$$m_{eff} = m_{veh} + \frac{4 J_{wheel} + J_{motor} N_{gear}^2}{r_{wheel}^2}$$

3.2 Electric Motor & Inverter Loss Model

Motor torque is constrained by the base-speed field-weakening envelope:

$$T_{max}(\omega) = \begin{cases} T_{peak} & \omega \le \omega_{base} \\ \frac{P_{peak}}{\omega} & \omega > \omega_{base} \end{cases}$$

Motor losses are decomposed into copper heating, iron hysteresis/eddy currents, and windage friction:

$$P_{loss}(\omega, T) = k_{cu} T^2 + k_{fe} \omega + k_w \omega^3 + P_{fixed}$$

3.3 High-Voltage Battery Equivalent Circuit Model (ECM)

The traction pack is modeled with an open-circuit voltage $V_{oc}(SOC)$ and internal resistance $R_{int}(SOC, T)$:

$$V_{terminal} = V_{oc}(SOC) - I_{batt} R_{int}(SOC, T_{batt})$$

$$I_{batt} = \frac{V_{oc} - \sqrt{V_{oc}^2 - 4 R_{int} P_{batt}}}{2 R_{int}}$$

$$SOC(t) = SOC_0 - \int_0^t \frac{I_{batt}(\tau)}{3600 \cdot Q_{capacity}} \, d\tau$$

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4. Proposed System Architecture

```
flowchart TD
    subgraph Data Layer
        GDir["Google Directions API"]
        GElev["Google Elevation API"]
        OffData["Offline Corridors Database"]
        VehDB["Indian BEV Database (10 Models)"]
    end

    subgraph Processing Layer
        REng["Route Engine (Smoothing, Grade, Traffic)"]
        DCGen["Driving Cycle Synthesizer (Eco, Normal, Perf)"]
    end

    subgraph Simulation Layer
        VVC["Virtual Vehicle Composer Interface"]
        SimBEV["Simulink BEV Model (BEV.slx)"]
        AnalytSolver["Fast Analytical Powertrain Solver"]
    end

    subgraph Optimization & UI Layer
        OptEngine["Charging & Mode Optimizer"]
        MapDash["Interactive Satellite/Roadmap UI"]
        PDFGen["Automated PDF Report Generator"]
    end

    GDir --> REng
    GElev --> REng
    OffData -.-> REng
    REng --> DCGen
    DCGen --> SimBEV
    DCGen --> AnalytSolver
    VehDB --> VVC
    VVC --> SimBEV
    VehDB --> AnalytSolver
    AnalytSolver --> OptEngine
    OptEngine --> MapDash
    MapDash --> PDFGen
```

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5. Work Breakdown Structure (WBS) & Task Allocation

Module / Subsystem	Primary Owner	Secondary Contributor	Key Deliverables
Powertrain Modeling & Dynamics	**Rajiv Gohil**	Sneha Pardeshi	Longitudinal forces equation, rotational inertia, Simscape/Simulink subsystem setup (`BEV.slx`).
Route Engine & Google Maps API	**Rajiv Gohil**	Sneha Pardeshi	Google Directions/Elevation REST client, offline fallback corridor generator, spatial interpolation.
Driving Cycle Synthesis	**Sneha Pardeshi**	Rajiv Gohil	2 Hz speed synthesizers, legal limit clamps, stop dwell models, Simulink `timeseries` exporter.
Energy Accounting & Costing	**Sneha Pardeshi**	Rajiv Gohil	Gross vs net kWh numerical integration, regen blending, HVAC auxiliary models, INR billing calculation.
Charging & Mode Optimization	**Sneha Pardeshi**	Rajiv Gohil	Reserve boundary monitor (*SOC < 15\%*), highway DC fast charger lookup, CC-CV taper duration algorithm.
Dashboard UI & Reporting	**Sneha Pardeshi**	Rajiv Gohil	App Designer UI, Google Maps satellite/roadmap tiles, Mapping Toolbox plotting, automated PDF generator.
Verification & Testing	**Rajiv Gohil**	Sneha Pardeshi	6 automated unit test suites (`runAllTests.m`), baseline numerical benchmarking.

6. Project Timeline & Milestone Schedule

```
gantt
    title BEV Intelligent Trip Planning Development Timeline
    dateFormat  YYYY-MM-DD
    section Phase 1: Planning & Architecture
    Problem Analysis & Requirements (LAB 6)   :done, 2026-08-10, 2026-08-23
    section Phase 2: Core Prototype Development
    Google Maps API & Route Engine Setup      :done, 2026-08-24, 2026-08-27
    Driving Cycle Synthesizer & Exporter      :done, 2026-08-25, 2026-08-28
    Working Prototype & Initial UI (LAB 7)    :done, 2026-08-26, 2026-08-29
    section Phase 3: Advanced Optimization
    Simulink BEV Model & Virtual Vehicle Prep :done, 2026-08-30, 2026-09-02
    Driving Modes & Charging Planner (LAB 8)  :done, 2026-09-01, 2026-09-05
    section Phase 4: Finalization & Evaluation
    Verification Testing & Slide Deck (LAB 9) :done, 2026-09-06, 2026-09-19
    Final Report & Submission Proof (LAB 10)  :done, 2026-09-20, 2026-09-26
```

7. Sign-off & Faculty Review

- **Student Signature (Lead):** Rajiv Gohil
- **Student Signature (Member):** Sneha Pardeshi
- **Faculty Guide / Reviewer:** Dept. of Electronics and Computer Science Engineering, VIT Mumbai

- **Review Status:** Accepted & Approved for Prototype Development